

Topic 7

Trigonometry

**Concept guide · Worked examples · Practice · Investigation and modelling
· Solutions****Learning focus**

Content explanations, original Cambridge-style practice and an investigation task tailored to International Mathematics 0607.

Topic map

7.1	Pythagoras theorem
7.2	Trigonometric ratios
7.3	Angles of elevation and bearings
7.4	Sine and cosine rules
7.5	Area of a non-right-angled triangle
7.6	Three-dimensional problems
7.7	Sensitivity to measurements

Core foundations and Extended applications are integrated. Practise exact methods, calculator work and the written reasoning required in investigations.

1. Right triangles

Pythagoras gives $\text{hypotenuse}^2 = \text{other side}^2 + \text{other side}^2$. The hypotenuse is opposite the right angle. For a missing shorter side, subtract squared lengths before taking a square root. In 3D, use one right triangle to find a face diagonal, then another for a space diagonal.

Worked example

A 6-8-right triangle has hypotenuse $\sqrt{6^2 + 8^2} = 10$.

Exam check: Check the longest side is the hypotenuse before substituting.

2. Sine, cosine and tangent

For a marked acute angle in a right triangle, $\sin = \text{opposite/hypotenuse}$, $\cos = \text{adjacent/hypotenuse}$, $\tan = \text{opposite/adjacent}$. Label sides relative to that angle before choosing. Use inverse trig for an angle and check the calculator is in degrees.

Worked example

Opposite = 5 and adjacent = 12, so $\theta = \tan^{-1}(5/12) = 22.6^\circ$ to one decimal place.

Exam check: Do not confuse an angle of elevation with the angle at the top of a vertical structure.

3. Non-right triangles

The sine rule links side/sine of its opposite angle for an opposite pair. The cosine rule $c^2 = a^2 + b^2 - 2ab \cos C$ uses the included angle C. The area formula is $\frac{1}{2} ab \sin C$ for two sides and the included angle. Sketch and label before selecting a formula.

Worked example

Sides 6 and 8 contain 30° . Area = $0.5 \times 6 \times 8 \times \sin 30^\circ = 12$ square units.

Exam check: With two sides and the included angle, cosine rule is usually the direct route to the third side.

4. Bearings, heights and 3D

Draw a plan view for bearings and a side elevation for height. In a cuboid, the face diagonal may form one side of a right triangle with vertical height. An angle between a line and a plane is measured against the line projection onto that plane.

Worked example

A cuboid has base 3 cm by 4 cm and height 12 cm. Base diagonal = 5 cm; space diagonal = $\sqrt{5^2 + 12^2} = 13$ cm.

Exam check: Show which triangle contains the required angle; a 3D sketch alone can hide the right angle.

5. Exact values and rounding

Recall $\sin 30^\circ = 1/2$, $\cos 60^\circ = 1/2$ and $\tan 45^\circ = 1$, plus the standard 0° , 30° , 45° , 60° , 90° values. Keep full calculator precision through intermediate stages, then round as directed. If a question requests an exact value, retain roots or π where appropriate.

Worked example

In a right triangle with hypotenuse 10 and angle 30° , the opposite side is $10 \sin 30^\circ = 5$.

Exam check: An intermediate rounded angle can noticeably change a later length.

6. Sensitivity to measurements

A model using an angle depends on both distance and angle accuracy. Repeat a trig calculation with a slightly changed angle to see how sensitive the prediction is; round at the end. The practical diagram can include a surveyor eye height if specified.

Worked example

At 30 m distance, height at 30° is $30 \tan 30^\circ \approx 17.3$ m, while at 32° it is ≈ 18.7 m.

Exam check: State model assumptions such as level ground.

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Graphical calculator and Paper 6

Set the calculator to degrees. Retain full precision during intermediate calculations, then round angles as instructed.

State the rule or model, display enough working to reproduce your result, test it against the given values and discuss at least one limitation.



Practice paper

Topic 7 · Trigonometry

Instructions

Original practice, informed by June 2026 components. Section A: short multiple choice for revision without a calculator. Section B: show full written working. Section C: investigation and modelling with reasons and checks.

Candidate name:

Class / date:

Total: 41 marks

Section A · Multiple choice

Choose one answer for each question.

1. Hypotenuse of sides 5 and 12? [1]

- A 11
- B 12
- C 13
- D 17

2. For opposite 3, hypotenuse 5, $\sin \theta$ is [1]

- A $\frac{3}{5}$
- B $\frac{4}{5}$
- C $\frac{5}{3}$
- D $\frac{3}{4}$

3. For adjacent 4, opposite 3, $\tan \theta$ is [1]

- A $\frac{4}{3}$
- B $\frac{3}{4}$
- C $\frac{3}{5}$
- D $\frac{4}{5}$

4. Which is the sine rule? [1]

- A $\frac{a}{\sin A} = \frac{b}{\sin B}$
- B $a^2 + b^2 = c^2$ for all triangles
- C $\sin A + \sin B = 1$
- D $a \cos A = b \cos B$

5. Area given a, b and included C is [1]

- A $ab \sin C$
- B $ab/2$
- C $ab \cos C/2$
- D $ab \sin C/2$

6. $\cos 60^\circ$ equals [1]

- A 0
- B $\frac{1}{2}$
- C $\frac{\sqrt{2}}{2}$
- D 1

7. If $\tan \theta = 1$ for an acute angle, θ is [1]

- A 30°
- B 45°
- C 60°
- D 90°

8. Cosine rule with included angle C gives $c^2 =$ [1]

- A $a^2 + b^2 + 2ab \cos C$
- B $a^2 + b^2 - 2ab \cos C$
- C $a^2 - b^2$
- D $ab \cos C$

Section B · Structured questions

9. A right triangle has legs 9 cm and 12 cm.

(a) Calculate the hypotenuse.

[2]

(b) Find the sine of the angle opposite the 9 cm side.

[2]

(c) Find the angle to one decimal place.

[2]

[Total: 6]

10. A vertical tower casts a 20 m shadow. The Sun elevation is 35° .

(a) Write an equation involving the tower height h .

[2]

(b) Calculate h to one decimal place.

[2]

(c) Explain why tangent is the appropriate ratio.

[2]

[Total: 6]

11. Triangle ABC has $AB = 8$ cm, $AC = 10$ cm and angle $BAC = 60^\circ$.

(a) Calculate BC using the cosine rule.

[3]

(b) Calculate its area.

[2]

(c) State the angle opposite BC.

[1]

[Total: 6]

12. A cuboid is 6 cm long, 8 cm wide and 24 cm high.

(a) Find the base diagonal.

[2]

(b) Find the space diagonal.

[2]

(c) Find the angle the space diagonal makes with the base, to one decimal place.

[3]

[Total: 7]

Section C · Investigation and modelling

Show the rule, test it with values and explain the result in context.

13. A surveyor measures a tower from a point 30 m away at an elevation of 30° .

(a) Write an expression for its height h . [2]

(b) Calculate h to 3 significant figures. [2]

(c) Calculate the model height if the angle were 32° . [2]

(d) Explain one reason a measured angle may change the prediction. [2]

[Total: 8]



Worked solutions

Fold back this section while completing the practice paper.

Marking routine

Method, model, communication and verification matter. A numerical answer should follow from clear working.

Section A · Answers and reasons

1. C — 13. Pythagoras gives $\sqrt{5^2+12^2} = 13$.
2. A — $3/5$. Sine relates opposite to hypotenuse.
3. B — $3/4$. Tangent relates opposite to adjacent.
4. A — $a/\sin A = b/\sin B$. Each side is divided by sine of its opposite angle.
5. D — $ab \sin C/2$. Triangle area for two sides and included angle is $ab \sin C/2$.
6. B — $1/2$. The exact cosine value at 60° is $1/2$.
7. B — 45° . The acute angle with equal opposite and adjacent sides is 45° .
8. B — $a^2+b^2-2ab \cos C$. The included angle is subtracted in $c^2 = a^2 + b^2 - 2ab \cos C$.

Section B • Worked solutions

9. a) $\sqrt{9^2+12^2}=15$ cm [2]. b) $9/15 = 3/5$ [2]. c) $\sin^{-1}(3/5) = 36.9^\circ$ [2].

10. a) $\tan 35^\circ = h/20$ [2]. b) $h = 20\tan 35^\circ = 14.0$ m [2]. c) Opposite height and adjacent shadow are known; tangent relates them [2].

11. a) $BC^2 = 8^2 + 10^2 - 2(8)(10)\cos 60^\circ = 84$; $BC = \sqrt{84} \approx 9.17$ cm [3]. b) $0.5(8)(10)\sin 60^\circ = 20\sqrt{3} \approx 34.6$ cm² [2]. c) Angle BAC [1].

12. a) $\sqrt{6^2 + 8^2} = 10$ cm [2]. b) $\sqrt{10^2 + 24^2} = 26$ cm [2]. c) $\tan \theta = 24/10$, $\theta = 67.4^\circ$ [3].

Section C • Investigation and modelling solutions

13(a). $h=30\tan 30^\circ$ [2].

13(b). $h\approx 17.3$ m [2].

13(c). $30\tan 32^\circ\approx 18.7$ m [2].

13(d). A 2° measurement error changes the result; the model also assumes level ground [2].

Self-check

Explain why your model fits the starting data, show an independent check and name a condition under which your prediction would be unreliable.

Reference: Cambridge IGCSE International Mathematics 0607 syllabus 2025-2027, <https://www.cambridgeinternational.org/Images/662472-2025-2027-syllabus.pdf>. Assessment conventions informed by the June 2026 papers and mark schemes supplied by the user. All practice questions are newly written.